Formula-Driven Diagrams

Objective

Read a formula-driven diagram whose geometry responds to time, then inspect a loop- and conditional-generated group whose membership stays fixed during playback.

Prerequisites

Complete lesson 07. No earlier
data mappings or modules are required to run this example.

Desktop workflow

In OpenAMX Desktop, open this document from `examples/`. Select **Run** to check its values, then **Preview** to inspect the mechanism and its playback controls. The diagram starts at the lower time bound; use play, pause, reset, scrub, and the optional loop control to inspect the crank-slider and generated trace points.

Inputs and support files

This lesson is input-free and imports no modules. Its geometry uses only ordinary AMX arithmetic, conditionals, `abs`, and `sqrt`.

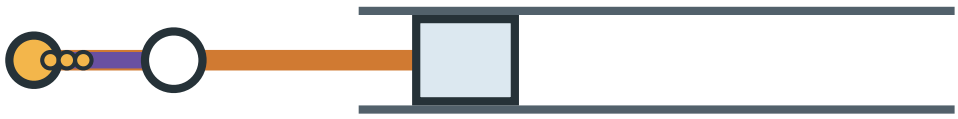
A crank-slider mechanism

```
let crankRadius: Number = 34
let rodLength: Number = 105
let pivotX: Number = 70
fn crankX(t: Number, r: Number): Number = r * (1 - 2 * abs(t - 1))
fn crankY(t: Number, r: Number): Number = if t <= 1 then sqrt(r * r - crankX(t, r) * crankX(t, r)) else -sqrt(r * r - crankX(t, r) * crankX(t, r))
fn sliderX(t: Number, pivot: Number, r: Number, rod: Number): Number = pivot + crankX(t, r) + sqrt(rod * rod - crankY(t, r) * crankY(t, r))
diagram crankSlider {
title: "Crank-slider mechanism"
description: "A crank pin follows one full circular path and drives a piston through a fixed-length connecting rod."
width: 320
height: 200
time t from 0 to 2 duration 4
line lowerGuide {
x1: 115
y1: 88
x2: 260
y2: 88
stroke: "#52616B"
lineWidth: 2
arrow: "none"
}
line upperGuide {
x1: 115
y1: 112
x2: 260
y2: 112
stroke: "#52616B"
lineWidth: 2
arrow: "none"
}
line connectingRod {
x1: pivotX + crankX(t, crankRadius)
y1: 100 + crankY(t, crankRadius)
x2: sliderX(t, pivotX, crankRadius, rodLength)
y2: 100
stroke: "#D07A32"
lineWidth: 5
arrow: "none"
```

```
}
line crank {
x1: pivotX
y1: 100
x2: pivotX + crankX(t, crankRadius)
y2: 100 + crankY(t, crankRadius)
stroke: "#6950A1"
lineWidth: 4
arrow: "none"
}
circle pivot {
cx: pivotX
cy: 100
r: 7
stroke: "#263238"
fill: "#FFFFFF"
lineWidth: 2
}
circle crankPin {
cx: pivotX + crankX(t, crankRadius)
cy: 100 + crankY(t, crankRadius)
r: 6
stroke: "#263238"
fill: "#F3B64B"
lineWidth: 2
}
rectangle piston {
x: sliderX(t, pivotX, crankRadius, rodLength) - 12
y: 90
width: 24
height: 20
stroke: "#263238"
fill: "#DCE8F0"
lineWidth: 2
}
text timeLabel {
x: 18
y: 24
value: "t = ${round(t, 2)} s"
fill: "#263238"
}
for sample in [0, 1, 2, 3] {
if sample > 0 {
circle tracePoint {
cx: pivotX + crankX(t, crankRadius) + sample * 4
cy: 100 + crankY(t, crankRadius)
r: 2
stroke: "#263238"
fill: "#F3B64B"
lineWidth: 1
}
}
}
}
```

Crank-slider mechanism

A crank pin follows one full circular path and drives a piston through a fixed-length connecting rod.



t = 0 s

A generated trace group

The loop visits four values and the conditional omits `sample == 0`, emitting three circles in source order. `tracePoint` is the one authored declaration repeated by the loop, not a unique source name for each emitted circle. The loop iterable and condition do not use `t`; the supported `cx` and `cy` properties do. The three generated points keep the same membership and order across frames while their positions follow the crank pin.

Expected outcome

The declared crank radius is 34 and the connecting-rod length is 105. Diagram-local t runs from 0 to 2 over a four-second playback duration. The loop and conditional emit exactly three trace circles when the diagram model is created. Their membership and order remain fixed during playback; their geometry, like the crank-slider geometry, is evaluated at each selected time. This is not a stateful solver or an accumulated simulation.

Standalone HTML and desktop Preview provide the interactive controls. PDF and DOCX exports contain a static frame at the lower time bound; neither includes playback controls. DOCX uses an image rather than editable native Word shapes.

Canvas coordinates are schematic coordinates, not physical units or engineering axes. The geometry uses no trigonometric function. No physical dimension, accessibility conformance, or screen-reader support is claimed. See the standalone battery-resistor circuit reference for a separate schematic and its stated interpretation.

Navigation

Index |
Previous: Tables, Charts, and Named Outputs |

Next: Runtime Type Checks